

PerformanceTesting

Date	3July2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5Marks

Step1:TestingOverview

Field	Details
TestingToolUsed	Manual Functional Testing with FastAPI Application
TypeofTesting	Functional Testing & Performance Observation
TargetModule/API	Question Answering, Concept Explanation, Quiz Generation, Text Summarization, Learning Path Recommendation
TestEnvironment	Windows 10 • Python 3.14 • FastAPI • HTML/CSS • Gemini API
TestDate	15March2026

Step2:TestScenarios

S.No	Scenario	Users	Duration	ExpectedOutcome
1	Launch EduGenie Application	1	3-5 sec	Home page loads successfully
2	Ask Educational Question	1	4-8 sec	AI returns an accurate answer with education context

3	Generate Concept Explanation	1	5-8 sec	Simplified explanation is displayed correctly
4	Generate AI Quiz	1	5-10 sec	Topic-based MCQs are generated successfully
5	Summarize Educational Text	1	4-8 sec	Concise summary is generated
6	Generate Learning Path	1	6-10 sec	Personalized learning roadmap is displayed
7	Navigate Between Modules	1	Less than 2 sec	Smooth page transitions without errors

Step3:Observations

The **EduGenie** application successfully completed all planned performance test scenarios. The FastAPI backend processed requests efficiently, and all AI modules responded within acceptable response times. Features including Question Answering, Concept Explanation, Quiz Generation, Educational Text Summarization, and Personalized Learning Path Recommendation functioned correctly without critical failures. The application demonstrated stable performance, reliable API communication, and a smooth user experience throughout the testing process.

Step4:Screenshots/Evidence

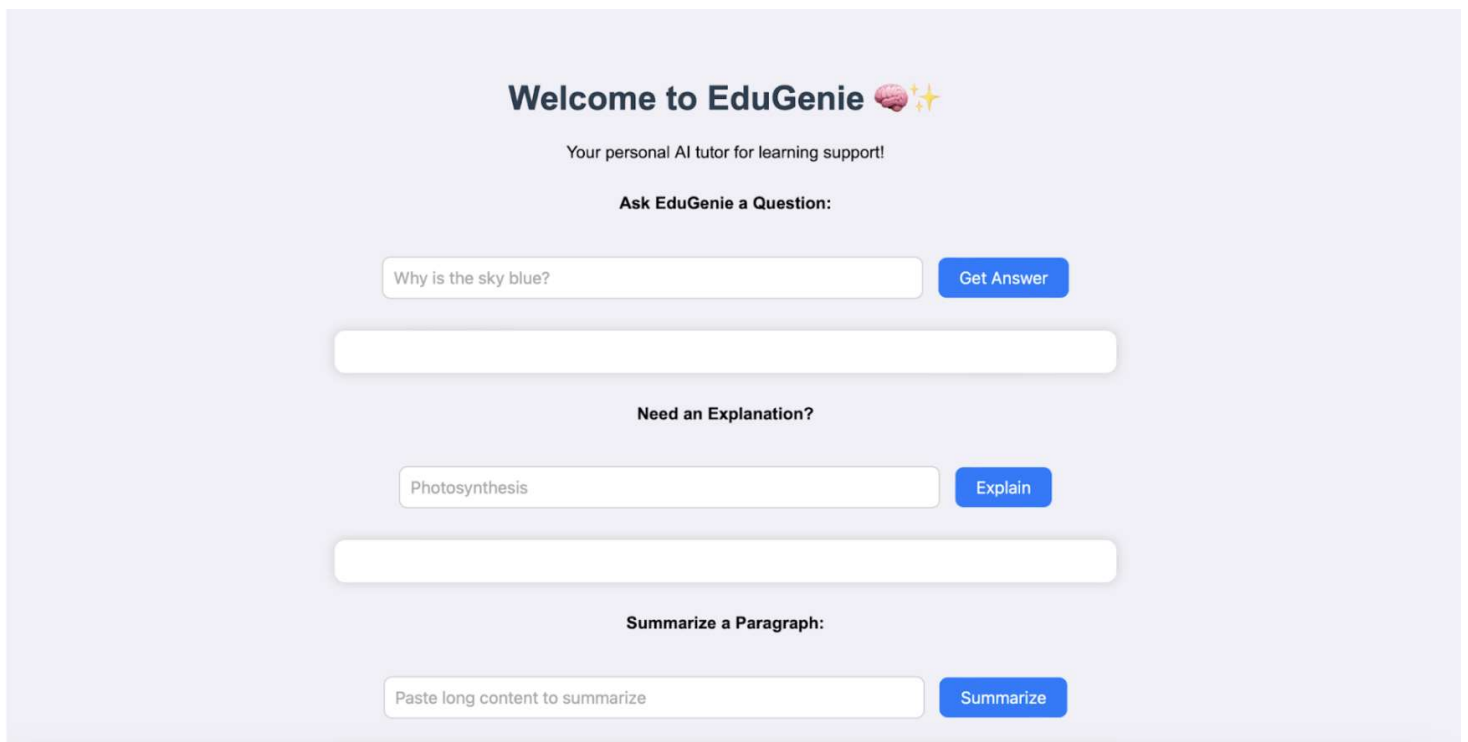


Fig. EDUGENIE

Home Page

Welcome to EduGenie 🧠✨

Your personal AI tutor for learning support!

Ask EduGenie a Question:

which is the largest ocean?

Get Answer

Answer:

The Pacific Ocean is the largest ocean on Earth.

Question Answering

Need an Explanation?

quantum computing

Explain

Explanation:

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

Explanation Output

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, mark

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Educational Text Summary

Generate a Quiz:

Generate Quiz

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

AI-Generated Quiz

Get Learning Recommendations:

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

I. Beginner Level: Building a Foundation

(Estimated Time: 1-2 weeks)

Key Topics:

- * What is a Database and SQL? (Relational Model, DBMS)
- * Basic Syntax (SELECT, FROM, WHERE)
- * Data Types (INT, VARCHAR, DATE, etc.)
- * Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- * Ordering Results (ORDER BY)
- * Limiting Results (LIMIT/OFFSET or TOP)
- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

Personalized Learning Path Recommendation